

The Integrated Supply and Demand Dashboards with Predictive Analytics for Oil Arbitrage and Price Forecasting

Yiwen (Elaine) Xu, Chien An Liu, Christian L Rodriguez, Kevin Zhu, Stephen Kim, Son Tran

This project aims to address a significant gap in the oil and gas trading space by consolidating global supply chain data into a single, user-friendly application that can support oil arbitrage decisions. Oil arbitrage is currently conducted through three main strategies: supply and demand analysis to take advantage of price differentials, price signals to take advantage of market patterns, and market sentiment to anticipate price movement. Current oil arbitrage practices are limited by the industry's resistance to adopt data analytic tools despite an overwhelming amount of available data and a reliance on disparate data sources which require manual effort to view together. The delay in adoption is a result of insufficient business support (Mohammadpoor & Torabi, 2020), cybersecurity concerns, system integration issues, and outdated infrastructure (Nguyen et al., 2020).

To overcome these challenges, a dashboard created merges data from relevant sources and the analytics based on the BEAST model gives more insights and better forecasts. It offers traders a centralized platform with integrated analysis tools for better decision-making, which is currently missing in the public domain. It also improves operational efficiency for commodities traders. The success can be measured by user adoption (if published) and the accuracy of its forecasts. There are two major risks associated with the project: Limits on data availability and difficulty in forecasting. The Energy Aspects API only provides data that dates back to 2009. Previous research highlighted various challenges of forecasting oil prices including disruptions from geopolitical events, natural disasters (Monge et al., 2023), and non-linear oil price reactions to refining and storage constraints (Kaufmann et al., 2008). Refineries are optimized for oil with specific viscosity and sulfur content, causing variable refining costs for oil of different qualities (Kaufmann et al., 2023).

The dashboard features BEAST and ARIMA models as proof of concept, offering key advantages over industry standards. Users can customize analyses to their needs, access centralized data seamlessly, and integrate custom models and visualizations thanks to its modular design. A user-friendly interface and enhanced visualization tools simplify navigation and interpretation, making complex data more accessible for meaningful insights.

For our visualization platform, Oil Lens opted for Power BI due to Microsoft's popularity among large enterprises and our testing found it to be more convenient and simpler to grasp compared to other visualization tools in the market like Tableau. For an industry that is a laggard in adopting new technologies, we believe this choice reduces the barrier to learning and adopting new technologies for employees in the oil and gas sector. If adopted in an enterprise environment, we could utilize Power BI's publishing features to deploy it to the Power BI Service. This deployment enables seamless sharing and collaboration across the team, allowing members to access up-to-date insights from any location. By setting appropriate permissions and access controls, we can ensure that sensitive data remains secure while empowering team members to make data-driven decisions efficiently.

The dashboard features data from all sectors of oil markets such as crude oil pricing, refining margins, refinery runs and maintenance schedules, oil production, global product inventories, imports/exports, and much more. Further, these fundamentals are organized in different tabs such as refining production and global stocks. All data was collected from Energy Aspects and the U.S.

Energy Information Administration by utilizing their APIs. Data in the dashboard features both macro and micro sectors of the industry. We began by importing and transforming raw data within Power BI, utilizing its robust data modeling capabilities to create meaningful relationships and calculations. With a focus on user experience, we designed interactive visuals and reports that allow users to explore data trends, patterns, and insights effortlessly. Given the complexity and interrelatedness of oil markets, an interactive dashboard and the ability to analyze the market at the macro and micro levels is imperative.

As illustrated in the Appendix, the dashboard features time series data for all relevant countries, geographical regions, and cartels in the industry, enabling users to drill down to a granular level such as covering production from all of a company’s assets (such as Suncor’s productive assets in Canada). Additionally, it showcases data on various crude oils globally, like Johan Sverdrup and Basrah, as well as major refined petroleum product classes such as gasoline, diesel, jet fuel, and fuel oil. Users can interact with these data points through simple clicks on the visualizations and intuitive filters and slicers, allowing them to customize their view according to specific interests or requirements such as products, countries, regions, time periods, etc.

As for integrating our forecasts into the dashboard, ARIMA and BEAST forecast outputs and respective accuracy metrics (such as MSE and MAPE) are available in different tabs. The group modeled 1123 time series datasets using BEAST and 1193 using ARIMA. As users interact with these dashboards, they can select the time series they’re interested in and the visualizations as well as the accuracy summary table will change accordingly.

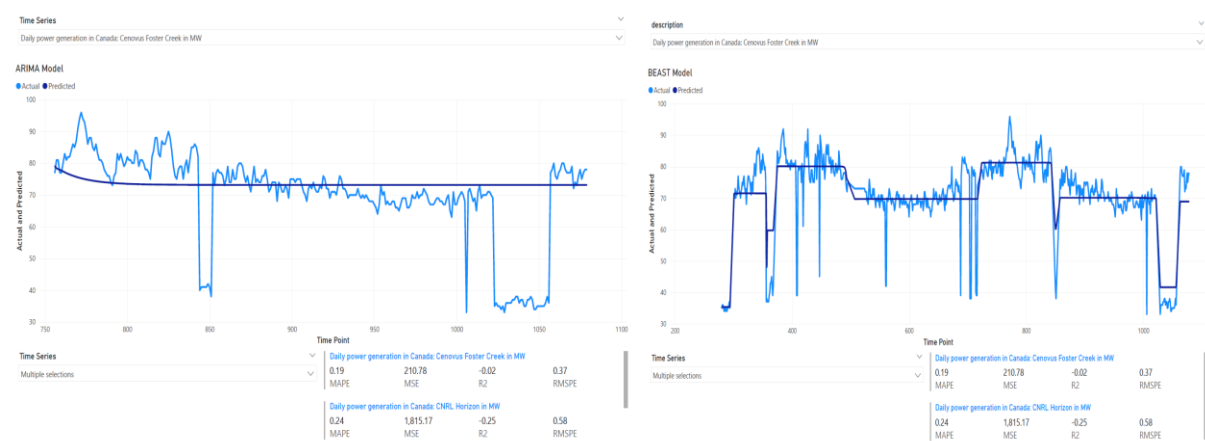


Figure 1: BEAST and ARIMA Forecasts from the Dashboard

One key innovation is the model choice for the computational component. The team has collected and processed over 6,000 time-series datasets on oil refining output, supply, and demand of crude oil and various crude products across various regions dating back to 2009. As the literature suggested, many factors, including but not limited to geopolitical risk (Kilian et al., 2024) and natural disasters (Monge et al., 2023) can cause frequent sharp changes to growth rates and volatility in commodity supply and demand. These sudden changes to commodity supply and demand can also greatly affect oil prices (Caldara et al., 2019). These fundamental changes to a time series dataset would require us to constantly retrain the models, or manually detect when a change occurred. To better model for these sharp changes, the team pivoted to a method that is capable of change detection as well as forecasting. We settled on a Bayesian change-point

detection and time series decomposition (BEAST) model to detect trend, noise, and seasonality, and critically identify stepwise changes in these features.

A Bayesian model like BEAST is suitable for detecting the change in a signal normally unobtainable from Non-Bayesian models (Gong, 2024). Compared to traditional methods which select a single “best” model for time-series analysis, BEAST embraces a Bayesian paradigm to integrate numerous competing models. Using these competing models, BEAST can be used to enhance the robustness of insights by quantifying uncertainties, addressing model misspecification, and reducing overfitting (Zhao et al., 2019). BEAST has also been shown to be useful for modeling the effects of market sentiment like social media (Reuning et al., 2022), war, or economic sanctions that can greatly affect oil pricing, supply, and demand (Benchimol et al., 2024). While current events and market sentiment are not covered in the scope of this project, BEAST was able to accurately fit a time series and detect fundamental changes in the dataset.

Given the reasons listed above, in our analyses, BEAST was used to improve forecast accuracy by identifying the latest subsets of continuous data points with the most recent trend from the datasets. Each subset is demarcated by the change points identified using BEAST. As there is not a consistent trend from supply and demand from different regions over the time horizon, it is a reasonable assumption that points following each changepoint would follow the current trend. For this reason, we detected each of the change points and identified the trends which would persist for the subsequent forecast period.

The project team started with evaluating the suitability of the BEAST model through backtesting: that is, the difference between actual and predicted values for the entire time series after fitting. While this may not be fully representative of actual forecasting accuracy, it is a suitable method of measuring if a model can fit patterns in the data. The results were excellent: BEAST is consistently able to achieve excellent back-test accuracy, typically extremely close to 1. This suggested that BEAST can estimate the trend and seasonality of our data and would likely produce reasonable forecasts. For illustration, the BEAST model was able to detect numerous sudden changes to crude oil inventories in China and achieve 0.99 R-squared in the back-test.

The project team then ran the BEAST model on the larger scale of datasets, and identified the change points from the sorted indices of all the change points. Since the data is time series, the regression model was trained on the period following the change points on a rolling basis based on all previous data points after the latest change point and used $N-1$ data points to detect trends, seasonality, and extract the latest trend to forecast the value for the future time point $t = N$. We compared the predicted values against the actual values. It turns out some of the data points are missing or incomplete, which is common in the oil inventory and production data collection, across the series, which misled the model training results. We replaced the datapoints far below the median with NaN, including nulls, and backward and forward filled the missing datapoints as a fair representative. Following replacing all invalid values with NaN, the results appear more accurate and meaningful (see appendix). The predicted values are approximately one hundred thousand across hundreds of data series.

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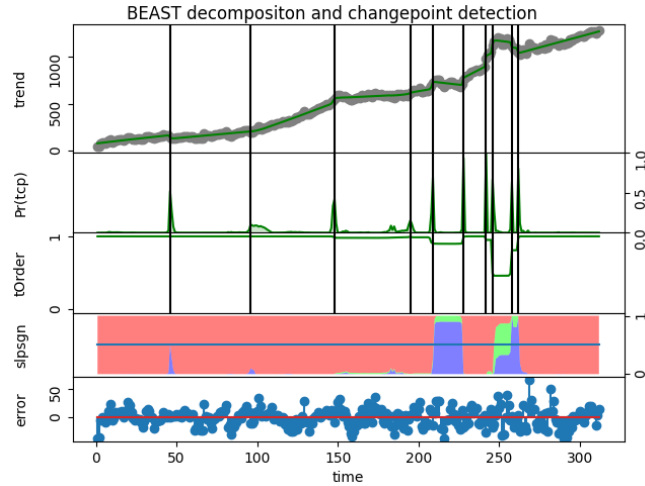


Figure 2: Monthly crude oil inventories in China from 2009 marked with periods detected by BEAST as well as its estimated trend and error across all periods.

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A proper evaluation of forecast accuracy is an important computational component of this study. The time-series cross-validation is empirically validated for performance estimation in time series (Cerqueira et al., 2020). For this study, a few forecast accuracy metrics were selected including R Squared, Mean Absolute Percentage Error (MAPE), Root Mean Squared Prediction Error (RMSPE), and Mean Squared Error (MSE) to evaluate the suitability of BEAST for the larger scale of data and ensure acceptable accuracy for generalization. To simply visualize the measurements, we grouped different data series by product categories of crude oil, oil products, NGLs, liquids, condensate, biofuels, and power, and plot the forecast accuracy metrics for each of the categories.

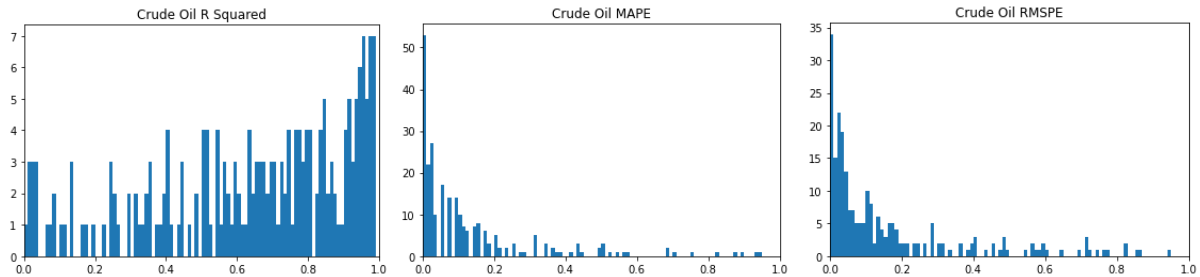


Figure 3: Accuracy Measures of BEAST Forecasts under Crude Oil

The performance of forecasts proved to be accurate across various forecast types (e.g. supply, demand, or regional forecast). In crude oil, the R Squared of data series above 0.6 takes up 67.9%, approximately 70%, and most of MAPE and RMSPE are between 0 and 0.2, indicating a very low error level.

In the category of Oil products, the percentage of R Squared higher than 0.6 is 63.6%, similar to crude oil, and MAPE and RMSPE both indicate lower levels of errors.

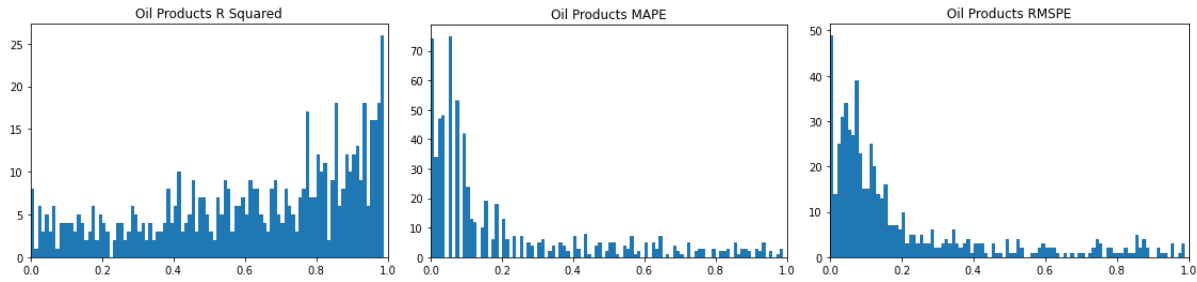


Figure 4: Accuracy Measure of BEAST Forecasts under Oil Products

The rest of product categories contain fewer data series relatively, however, the majority of these series demonstrate strong predictive performance, with R Squared value exceeding 0.6. Most error percentages, as indicated by MAPE and RMSPE, fall within the range of 0 and 0.2, reflecting high accuracy and reliability in predictions. The charts are included in the appendix.

The second algorithm employed in this project was the ARIMA model, since an ARIMA model is superior to short-term forecasting (Kim, 2018). Forecasting was performed using a 70/30 split of the data available. Two approaches were tested: the basic ARIMA model, with predefined parameters for the order of autoregressors, differencing, and moving-averages components (p , d , q) set as (1, 1, 1), and the auto-ARIMA model from the `pmdarima` library. While the auto-ARIMA model is computationally more expensive, as it identifies the optimal (p , d , q) parameters by minimizing the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) and settling on a single-fitted ARIMA model for multiple time series forecasting, it offers greater accuracy, effectiveness, and scalability. These attributes make it particularly suitable for forecasting thousands of time series that come from a combination of supply and demand chains.

To manage computational costs, we compromised by employing a static data-splitting approach rather than a rolling window calculation, which requires significantly higher computational resources by approximately 100x. This can be developed further by future studies. Data preprocessing for the ARIMA models followed the same preprocessing procedures as those applied for the BEAST analyses. The auto-ARIMA models were then trained on all available time series, selecting the best models/parameters for forecasting. This analysis provided valuable insight into how traditional forecasting methods perform compared to those using BEAST. Accuracy metrics such as MAPE, RMSPE, and R-squared were used to evaluate the model performance and were visualized through histograms organized by product category. Some product categories included much longer data series, while others contained far fewer.

The forecasting performance of ARIMA models was less favorable than expected when compared to BEAST models. For example, the MAPE and RMSPE values for different series from crude oil were relatively high, although most remained between 0 and 0.4. R-squared values, mostly below 0.3, were significantly lower than those achieved with BEAST forecasts.

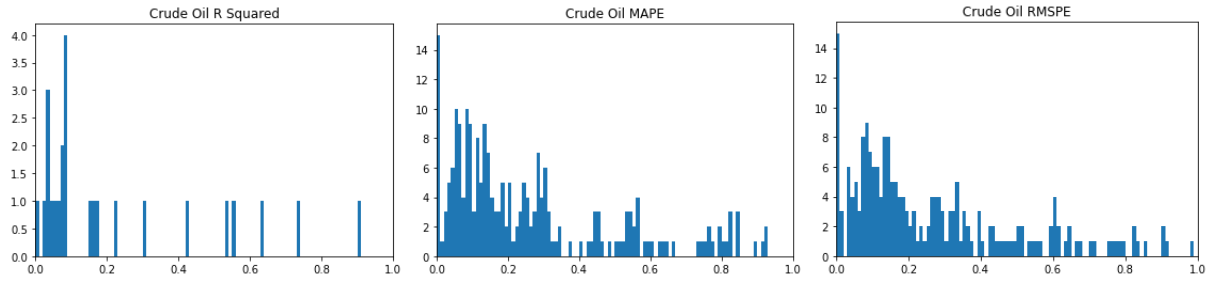


Figure 5: Accuracy Measures of ARIMA Forecasts under Crude Oil

Similar patterns were observed in oil products, where MAPE and RMSPE are more evenly distributed across values between 0 and 1, but lower R-squared values remained lower.

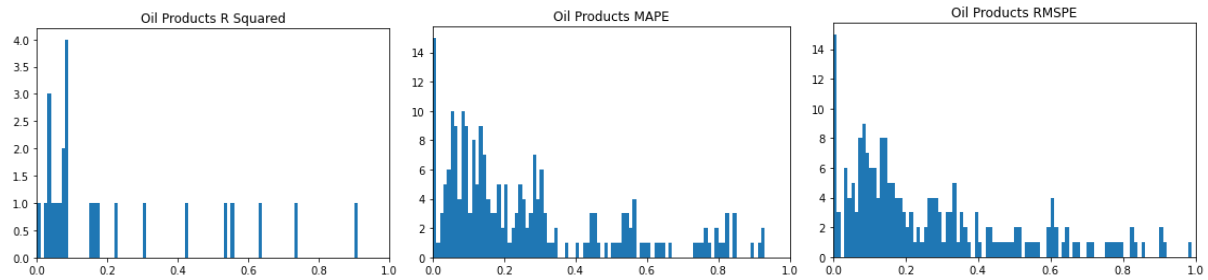


Figure 6: Accuracy Measure of ARIMA Forecasts under Oil Products

These observations demonstrate that the BEAST models we selected and our forecasting power outperform the traditional methods, especially when considering data for oil and gas supply and demand chains.

Walk-forward validation was used as cross-validation to evaluate ARIMA models. To accomplish this, we created a method named `get_accuracy` that returns an RMSE value from a time series. The method takes the time series drops all N/A values, and then splits the time series into a training set and a test set. A point from the test set is appended to the training set and an ARIMA model is trained using the new training dataset. This ARIMA model is then used to forecast a value which is appended to a list of predictions. This is repeated using a loop until all test points have been appended to the training set. The root mean square error equation is then applied to the list of predictions and test points. This experiment demonstrates how ARIMA performs when applied to each named time series in your dataset. Each RMSE value is stored in a dictionary with the time series name used as a key. The experiment also tests for overfitting and creates a metric for how the models will perform in a more real world scenario through cross validation.

In conclusion, the team was able to innovate on creating an Oil Arbitrage Dashboard that integrates Supply and Demand, augmented with Predicted Analytics and Price Forecasting. We processed over 6,000 time-series datasets on oil refining output and ran BEAST and ARIMA models to get time-series forecasting for oil products. Using PowerBI, we created the dashboard which incorporated the predictions and data points for Supply and Demand in a user-friendly way resulting in a cutting-edge solution. All team members have contributed a similar amount of effort.

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Appendix A

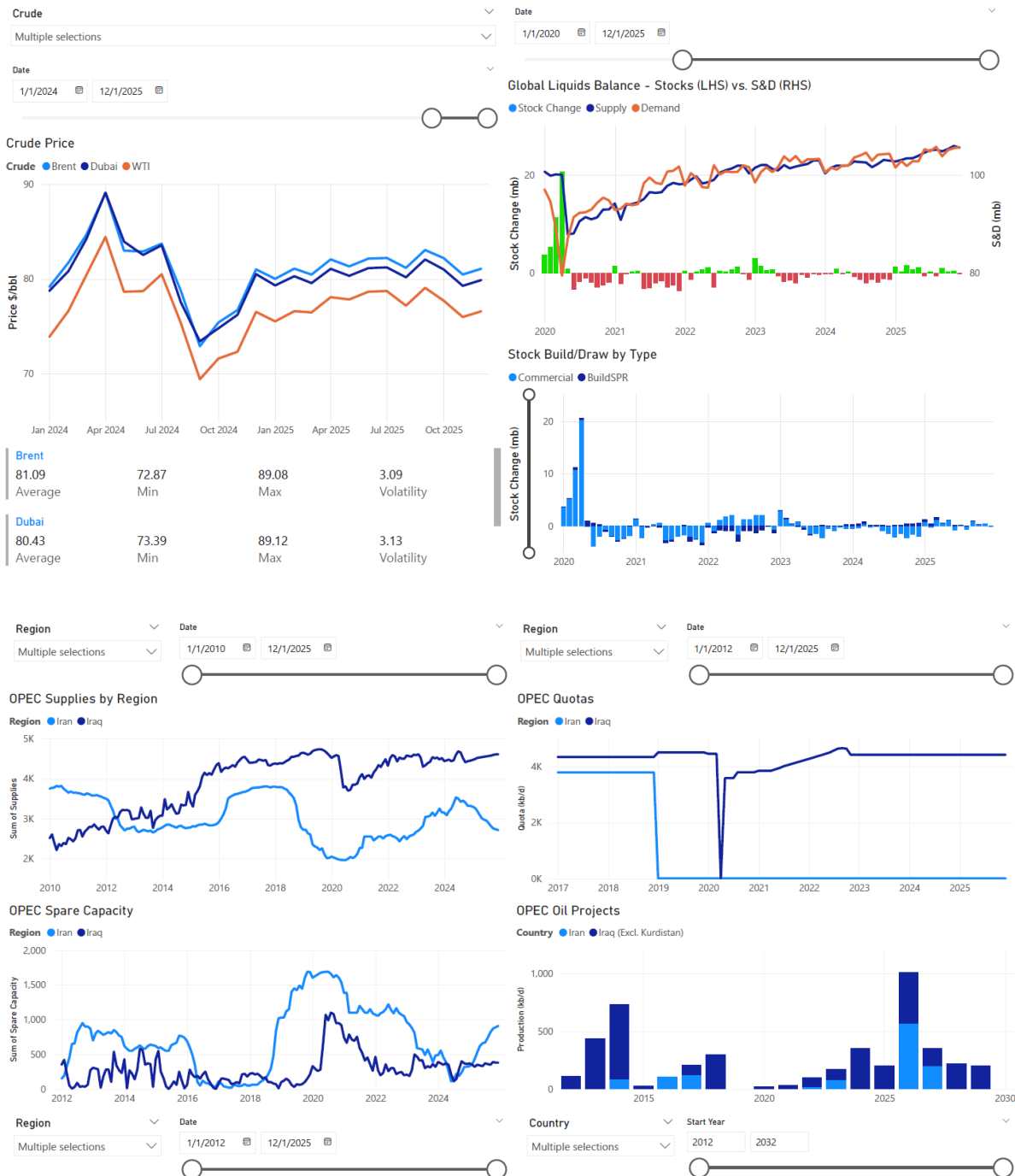


Figure A1: Examples of Dashboards

Appendix B

Results after removing nulls

	real	pred		real	pred
0	8532.00	8286.806306	0	8532.00	8891.463574
1	0.00	0.000000	1	11622.00	12406.000000
2	46565.00	48000.857143	2	46565.00	48000.857143
3	2558.34	2852.828700	3	2558.34	2852.828700
4	56648.16	50531.030649	4	56648.16	51583.095306
...	...	...	...	...	...
96	0.00	0.000000	96	27080.00	32144.000000
97	93050.00	95250.333333	97	93050.00	95250.333333
98	0.00	7822.470127	98	8584.00	7978.727106
99	0.00	0.000000	99	2777.00	2717.022059
100	24720.00	23772.357143	100	24720.00	15078.656025



Appendix C Accuracy Measures for Other Categories

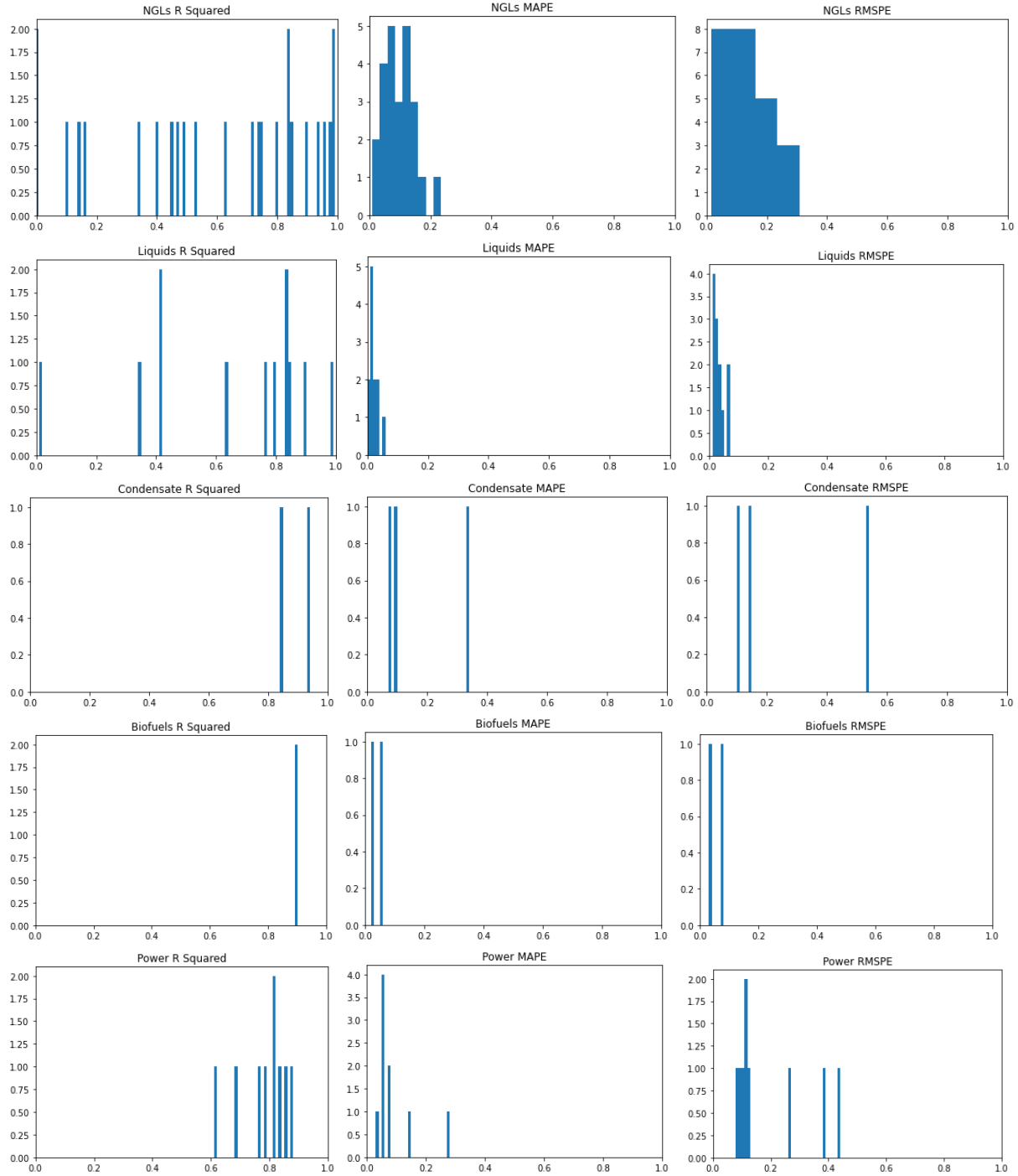


Figure C1: Accuracy Measures of BEAST Forecasts

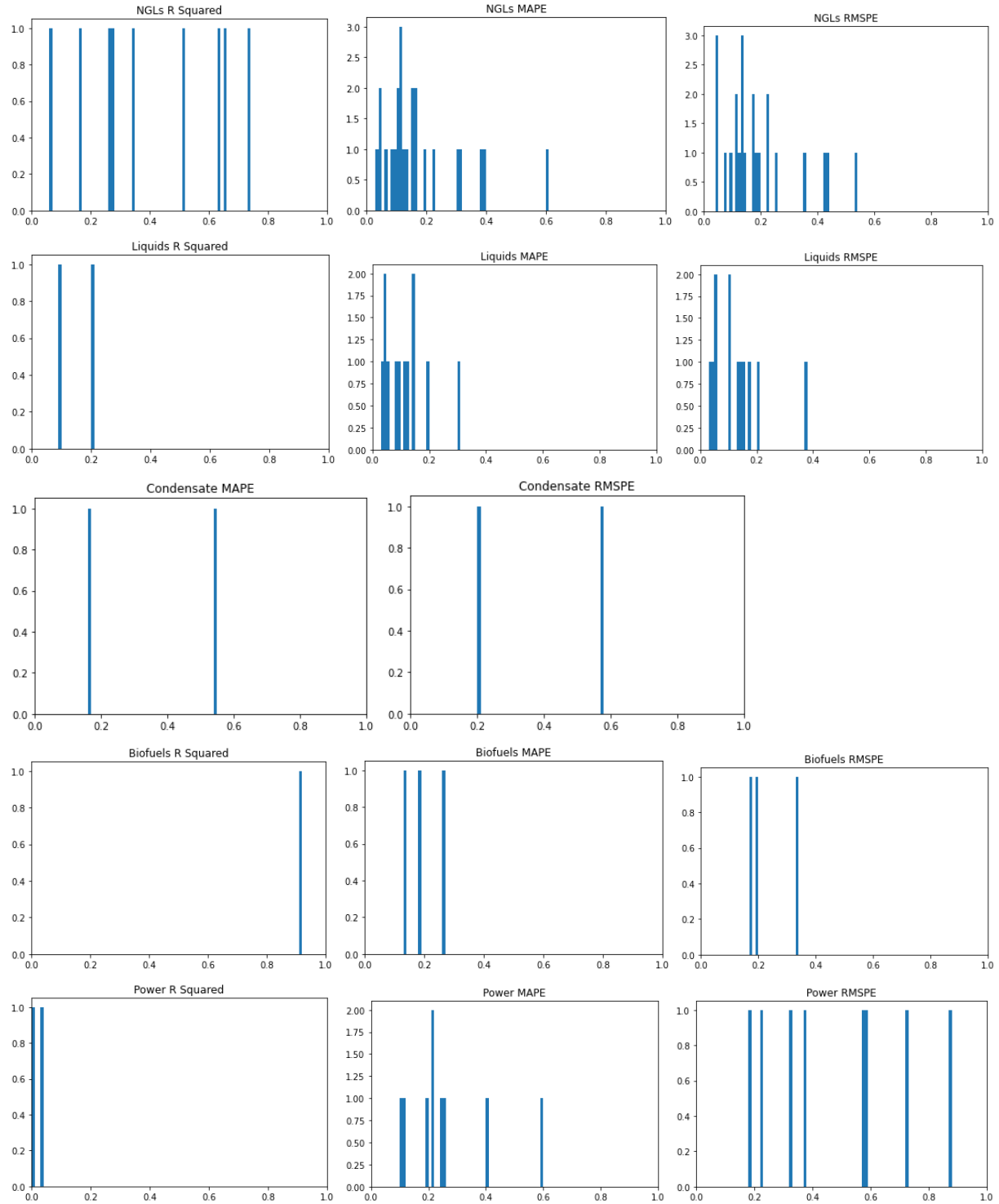


Figure C2: Accuracy Measures of ARIMA Forecasts